

Mutual simulation of Hamiltonian dynamics on interacting quantum systems

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Control-theoretical model of quantum computers

Time evolution defined by the time-dependent Schrödinger equation

$$\frac{d}{dt}U(t) = -iH(t)U(t), \quad U(0) = \mathbf{1}$$

$H(t)$ Hamiltonian

$U(t)$ is the resulting unitary after time t

$$H(t) = H_d + \sum_{i=1}^m a_i(t)H_i$$

H_d drift Hamiltonian internal to the system

$\sum_{i=1}^m a_i(t)H_i$ control Hamiltonian externally changeable

Complexity of a unitary U : minimal time required to implement U

Fast control limit

strength of the control Hamiltonians arbitrarily large $\implies$

control unitaries in control group K arbitrarily fast

Complexity of a unitary U : minimal time

$$U = \exp(-iHt_N)V_N \cdots \exp(-iHt_2)V_2 \exp(-iHt_1)V_1$$

express it as the evolution according to the piecewise constant Hamiltonian

$$U = U_N \exp(-iU_N^\dagger H U_N t_N) \cdots \exp(-iU_2^\dagger H U_2 t_2) \exp(-iU_1^\dagger H U_1 t_1)$$

Average Hamiltonian Theory

Dyson expansion

$$U(t) = \mathcal{T} \exp\left(\int_{\tau=0}^t d\tau H(\tau)\right) = \exp(-i\bar{H}t)$$

Magnus expansion $\bar{H} = \bar{H}^{(0)} + \bar{H}^{(1)} + \dots$

$$\bar{H}^{(0)} = \frac{1}{t} \int_0^t d\tau H(\tau), \quad \|\bar{H}^{(0)}\| \leq 1$$

$$\bar{H}^{(1)} = \frac{-i}{2t} \int_0^t d\tau' \int_0^t \tau'' [H(\tau'), H(\tau'')], \quad \|\bar{H}^{(1)}\| \leq t/2$$

$\implies U(t)$ essentially determined by $\bar{H}^{(0)}$ for small t

Simulating Hamiltonians (first order simulation)

H can simulate $\tilde{H}$ with overhead 1, written $\tilde{H} \leq H$, and N time steps iff

$$\tilde{H} = \sum_{j=1}^N p_j U_j^\dagger H U_j,$$

where $p_1, \dots, p_N$ are positive real numbers summing to 1 and $U_j \in K$ are control operations, i.e. $\tilde{H}$ is a convex sum of unitary conjugates of H

H can simulate $\tilde{H}$ with overhead τ iff $\tilde{H} \leq \tau H$

time-optimal simulation problem $\longrightarrow$ convex optimization problem

Composite systems

system consisting of n qudits $\mathcal{H} = \mathbb{C}^d \otimes \mathbb{C}^d \otimes \dots \otimes \mathbb{C}^d$

pair-interaction Hamiltonian

$$H = \sum_{k < l} \sum_{\alpha \beta} J_{kl; \alpha \beta} \sigma_{\alpha}^{(k)} \sigma_{\beta}^{(l)} + \sum_k \sum_{\alpha} r_{k; \alpha} \sigma_{\alpha}^{(k)},$$

where $B := \{\sigma_{\alpha} \mid \alpha = 1, \dots, d^2 - 1\}$ is a basis of $su(d)$

control operations are all local operations in

$$K := SU(d) \otimes SU(d) \otimes \dots \otimes SU(d)$$

Decoupling/Selective decoupling

quantum computing requires the ability to perform gates

couplings created by gates can only originate from natural couplings

naturally available Hamiltonians do not couple specific pairs of qubits

→ turn off disturbing couplings

Time-reversal

evolution of a single spin seems is phenomenologically a relaxation/
dephasing process

however, it is not always irreversible

→ refocusing simulate $-H$ when H is the system Hamiltonian

Universal simulation of quantum dynamics

simulating quantum dynamics on a classical computer is hard

→ universal quantum simulator

use a quantum computer to simulate arbitrary dynamics

Algebraic Graph Theory gives Lower Bounds on Complexity

represent pair-interaction Hamiltonian

$$H = \sum_{k < l} \sum_{\alpha \beta} J_{kl; \alpha \beta} \sigma_{\alpha}^{(k)} \sigma_{\beta}^{(l)} + \sum_k \sum_{\alpha} r_{k; \alpha} \sigma_{\alpha}^{(k)},$$

where $B := \{\sigma_{\alpha} \mid \alpha = 1, \dots, d^2 - 1\}$ is a ONB of $su(d)$

as coupling matrix (extension of adjacency matrix of a graph)

$$J = \left(\begin{array}{c|c|c|c|c} 0 & J_{12} & J_{13} & \cdots & J_{1n} \\ \hline J_{21} & 0 & J_{23} & \cdots & J_{2n} \\ \hline \vdots & \vdots & & & \\ \hline J_{n1} & J_{n2} & J_{n3} & & 0 \end{array} \right) \in \mathbb{R}^{(d^2-1)n \times (d^2-1)n}$$

Analogy: coupling matrix $\leftrightarrow$ adjacency matrix

consider dipole-dipole coupling between all nodes

$$H = \sum_{k < l} w_{kl} (3 \sigma_z \otimes \sigma_z - \sum_{\alpha=x,y,z} \sigma_\alpha \otimes \sigma_\alpha), \quad w_{kl} \neq 0$$

coupling matrix can be expressed as

$$J = W \otimes C,$$

where $W = (w_{kl})$ and $C = \text{diag}(-1, -1, 2)$.

W adjacency matrix of a weighted graph

C characterizes the type of coupling between nodes

Action of control operations on the coupling matrix J

Conjugation of the pair-interaction Hamiltonian H by control operations

corresponds to

conjugation of J by a direct sum of orthogonal matrices

$$H \mapsto (U_1 \otimes U_2 \otimes \cdots \otimes U_n)^\dagger H (U_1 \otimes U_2 \otimes \cdots \otimes U_n)$$

$$J \mapsto (O_1 \oplus O_2 \oplus \cdots \oplus O_n) J (O_1 \oplus O_2 \oplus \cdots \oplus O_n)^T$$

Lower bounds on time overhead

If H can simulate $\tilde{H}$ with overhead τ then

$$\tilde{J}/\tau = \sum_{j=1}^N p_j (U_{j1} \oplus U_{j2} \oplus \cdots \oplus U_{jn}) J (U_{j1} \oplus U_{j2} \oplus \cdots \oplus U_{jn})^T$$

convex sum of conjugates of J

$\Rightarrow$ necessary condition given by [majorization criterion](#):

the spectrum $\text{Spec}(\tilde{J})$ majorizes the spectrum $\text{Spec}(\tau J)$

Application I: Time reversal

Time reversal: $H \mapsto -H$ (refocusing, spin echoes)

dipole-dipole coupling between all nodes (easy to invert)

$$\begin{aligned} H &= \sum_{k < l} (3 \sigma_z \otimes \sigma_z - \sum_{\alpha=x,y,z} \sigma_\alpha \otimes \sigma_\alpha) \\ J &= K \otimes \text{diag}(-1, -1, 2), \end{aligned}$$

where K is the adjacency matrix of the complete graph

lower bound on time overhead is 2

this bound is tight (well known-scheme in NMR)

Application I: Time reversal

strong scalar coupling between all nodes (difficult to invert)

$$\begin{aligned} H &= \sum_{k < l} (\sigma_x^{(k)} \sigma_x^{(l)} + \sigma_y^{(k)} \sigma_y^{(l)} + \sigma_z^{(k)} \sigma_z^{(l)}) \\ J &= K \otimes \mathbf{1}_3 \end{aligned}$$

lower bound on the overhead $n - 1$

the inverted time-evolution is necessarily slower by the factor $(n - 1)$

upper bound $cn - 1$ (decoupling scheme based on orthogonal arrays)

Decoupling and time-reversal schemes (Upper bounds)

Conjugating by unitaries of an **error basis** $\mathcal{E} = \{U_1 = \mathbf{1}, U_2, \dots, U_{d^2}\}$

$$\sum_j U_j^\dagger a U_j = \mathbf{0} \quad \text{for all } a \in su(d)$$

cancels all matrices in $su(d)$

Combinatorial problem: which sequences of conjugations allow to switch off all interactions?

Efficient solution: decoupling in networks based on **orthogonal arrays**

number of conjugations grows linearly with n

Application II: Hierarchy of Hamiltonians

$$H_1 = \sum_{k < l} w_{kl} (\sigma_x^{(k)} \otimes \sigma_x^{(l)} + \sigma_y^{(k)} \otimes \sigma_y^{(l)})$$

$$H_2 = \sum_{k < l} w_{kl} (\sigma_x^{(k)} \otimes \sigma_x^{(l)} - \sigma_y^{(k)} \otimes \sigma_y^{(l)})$$

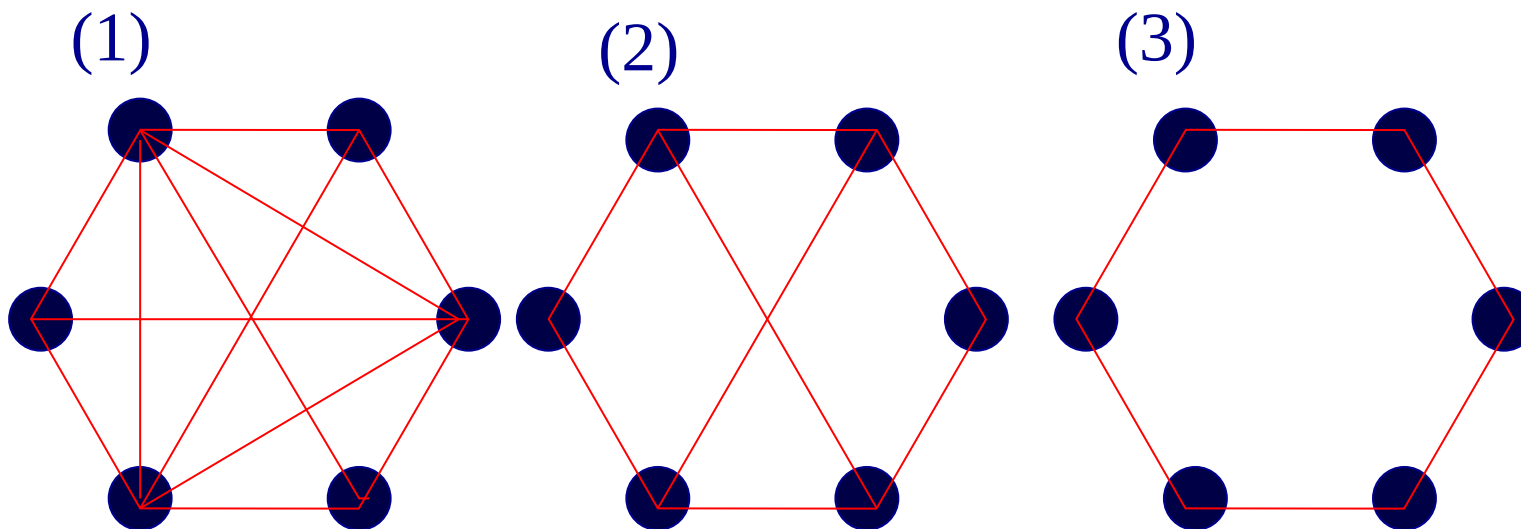
H_1 cannot simulate H_2 with overhead less than $(n - 1)$

H_2 can simulate H_1 with overhead 2

Application III: Switching off **some** interactions

Combinatorial problem: cancel some interactions

Are the other interactions necessarily disturbed?



Yes, for some graphs $\rightarrow$ lower bound on time overhead given by eigenvalues of the adjacency matrix

Upper bound given by chromatic index of interaction graph

Universal simulation using finite control groups

Universal simulation does not require all operations in $SU(d)$

Certain subgroups are sufficient; called transformer groups

Finite transformer groups are characterized in terms of group characters

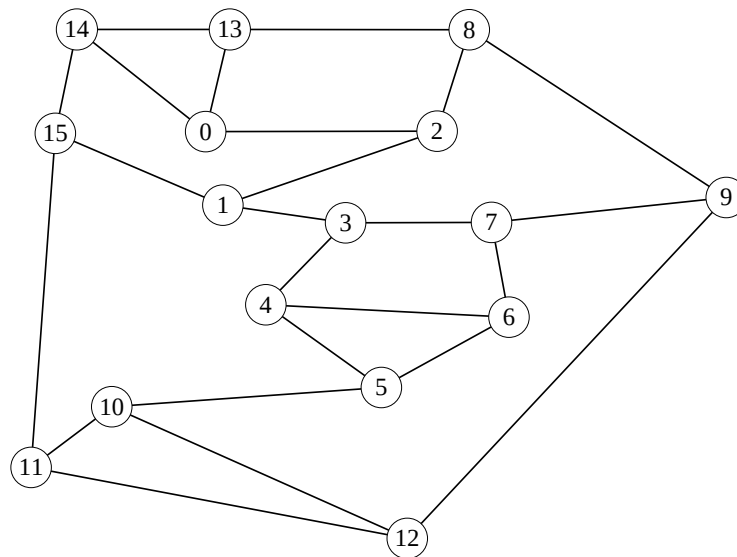
Treating the Independent Set Problem by 2D Ising interactions with Adiabatic Quantum Computing

construct a “physically reasonable” Hamiltonian whose ground states encode the solutions of Independent Set:

- INSTANCE: Graph $G = (V, E)$, positive integer $v \leq |V|$.
- QUESTION: Does G contain an independent set whose cardinality is at least v , i.e., a subset $V' \subseteq V$ such that $|V'| \geq v$ and such that no two vertices in V' are joined by an edge in E ?

Independent Set remains NP-complete for cubic planar graphs.

Cubic planar graph



Theorem (Planar spin glass within a magnetic field)

Let $G = (V, E)$ be a cubic planar. Determining the energy of the ground states of the corresponding Hamiltonian

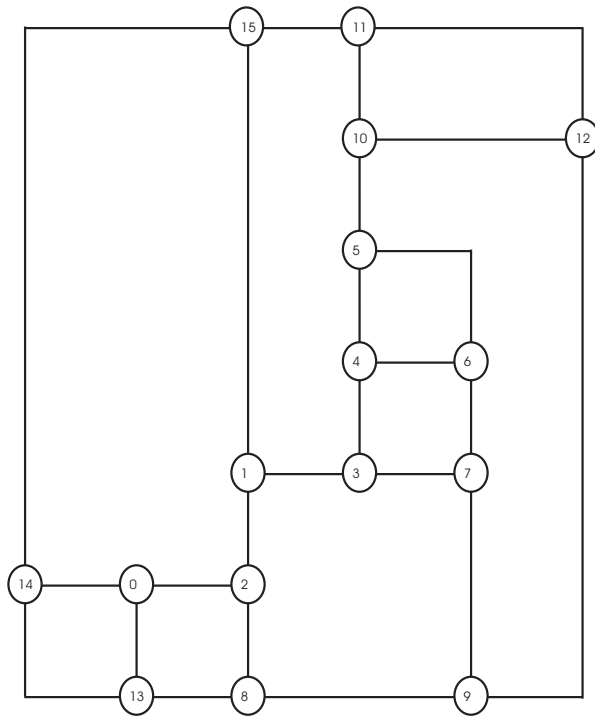
$$H_P = \sum_{(k,l) \in E} \sigma_z^{(k)} \sigma_z^{(l)} + \sum_{k \in V} \sigma_z^{(k)} \quad (1)$$

is equivalent to determining the maximum cardinality of independent sets of G .

Problem: H_P is **not physically reasonable**

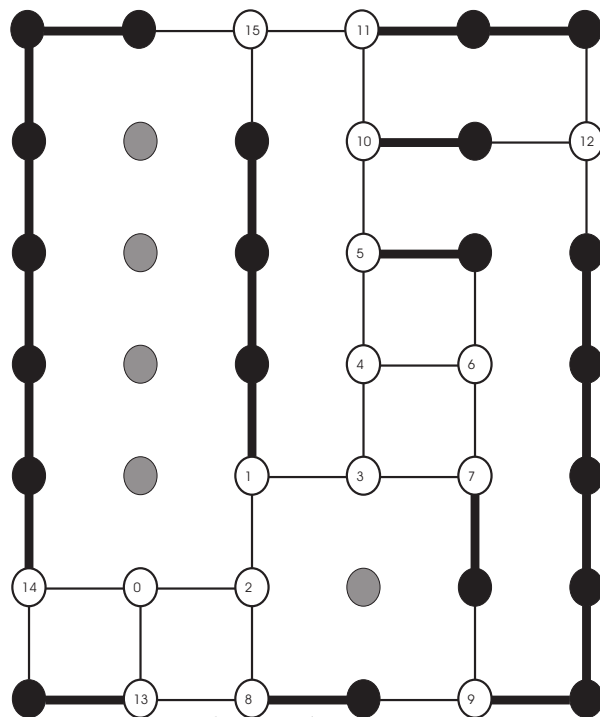
Planar orthogonal embedding of a graph $G = (V, E)$ maps

- vertices $k \in V$ to lattice points $\Gamma(k)$ and
- edges $(k, l) \in E$ to paths in the lattice such that the images of their endpoints $\Gamma(k)$ and $\Gamma(l)$ are connected and such that the paths do not share any vertices (besides the endpoints).



Planar orthogonal Hamiltonians

a physically reasonable Hamiltonian whose ground states correspond 1-to-1 to the ground states of H_P



filled circles: dummies, no local σ_z terms

thick lines: $-\textcolor{red}{c}\sigma_z \otimes \sigma_z$ coupling

circles: vertices $\Gamma(k)$, σ_z as local term

thin lines: $\sigma_z \otimes \sigma_z$ coupling

Mathematical tools

Vector majorization

let $x = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$

define $x^\downarrow = (x_1^\downarrow, x_2^\downarrow, \dots, x_d^\downarrow) \in \mathbb{R}^d$ as the vector whose components are the same as of x , but ordered so that

$$x_1^\downarrow \geq x_2^\downarrow \geq \dots \geq x_d^\downarrow$$

x majorizes y , written $x \preceq y$, if

$$\sum_{j=1}^k x_j^\downarrow \leq \sum_{j=1}^k y_j^\downarrow$$

for $k = 1, \dots, d-1$ and

$$\sum_{j=1}^d x_j^\downarrow = \sum_{j=1}^d y_j^\downarrow$$

Mathematical tools

Operator majorization

let A and B be two Hermitian matrices of size d

we say A majorizes B , written $A \preceq B$, if

$$\lambda(A) \preceq \lambda(B),$$

where $\lambda(A)$ and $\lambda(B)$ are the spectra (the vectors of eigenvalues) of A and B , respectively.

Uhlmann's theorem

we have $A \preceq B$ iff

$$A = \sum_{j=1}^N p_j U_j^\dagger B U_j,$$

where $\{p_j\}$ is a probability distribution and $\{U_j\}$ is a set of unitaries